

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Ramamoorthy

QUESTIONS: 12

Total Marks:60

Annexure A

CONCEPT MAPPING

Code of Conduct:

I _Pavithra Sri .S (192571079)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pavithra Sri .S

CONCEPT MAPPING

1. Construct a Concept Map

Given:

Requirement Analysis → _____ → Coding → Testing → _____ → Maintenance

Answer:

Design, Deployment

Explanation

The Design phase transforms the requirements gathered during analysis into a detailed software blueprint that defines the system architecture, database structure, class relationships, interfaces, and workflows. It provides developers with a clear implementation strategy. Deployment is the final phase where the tested software is installed in the production environment and made available to end users. It includes installation, configuration, user training, and performance monitoring to ensure successful software adoption.

Explanation & Linkages

Within the SDLC, Design serves as the foundation for all development activities by converting user requirements into technical specifications. Coding and testing are performed according to the design documents. After successful verification, Deployment delivers the validated software to the operational environment. Every phase depends on the successful completion of the previous one, creating a systematic and controlled development process.

Analysis

A well-planned design significantly reduces implementation errors and simplifies testing activities. Proper deployment ensures software reliability, minimizes operational risks, and improves customer satisfaction. Together, these phases enhance software quality, maintainability, and long-term system performance.

(5 Marks)

2. Construct a Concept Map

Given:

Object-Oriented Principles

→ Abstraction

→ _____

→ Inheritance

→ _____

Answer:

Identified Concepts

Encapsulation, Polymorphism

Explanation

Encapsulation protects an object's internal data by restricting direct access and allowing interaction only through well-defined methods. Polymorphism enables a single interface or method to perform multiple behaviors depending on the object invoking it, thereby supporting runtime flexibility.

Explanation & Linkages

Object-Oriented Design is built upon Abstraction, Encapsulation, Inheritance, and Polymorphism. Encapsulation improves data security by combining attributes and methods within a single class. Polymorphism works with inheritance to allow different subclasses to implement common methods differently, increasing software flexibility without changing the interface.

Analysis

These principles reduce code duplication, improve software security, encourage code reuse, and simplify maintenance. They enable developers to build modular, scalable, and reusable applications capable of adapting to changing requirements.

(5 Marks)

3. Construct a Concept Map

Given:

Class → Blueprint

Object → Instance

Method → _____

Attribute → _____

Answer:

Identified Concepts

Behaviour, State

Enhanced Explanation

A Class is a blueprint that defines the properties and operations of an object. An Object is an instance of that class. The State of an object is represented by its attributes, while its Behaviour is represented by methods that perform specific operations.

Elaborated Explanation & Linkages

Each object stores its own state while sharing the behaviours defined by its class. Methods manipulate attribute values, allowing objects to interact with one another through message passing. This relationship accurately models real-world entities within software systems.

Enhanced Analysis

Separating state and behaviour improves modularity, readability, and maintainability. It promotes better software organization and enables developers to create reusable and extensible object-oriented applications.

(5 Marks)

4. A Concept Map Shows

Given:

Software Quality

→ Reliability

→ _____

→ Maintainability

Answer:

Identified Concepts

Usability

Explanation

Usability measures how effectively and efficiently users can interact with a software system. A usable application is intuitive, easy to learn, and minimizes user errors while improving overall productivity.

Explanation & Linkages

Software quality depends on Reliability, Usability, Maintainability, Efficiency, and Security. Reliability ensures consistent operation, while Maintainability supports future modifications. Usability connects these attributes by ensuring that technical excellence translates into a positive user experience.

Analysis

High usability improves customer satisfaction, reduces training requirements, increases software acceptance, and contributes significantly to the commercial success of software products.

(5 Marks)

5. UML Concept Map

Given:

UML

→ Structural Diagrams → _____

→ Behavioral Diagrams → Activity Diagram

→ Interaction Diagrams → _____

Answer:

Identified Concepts

Object Diagram, Communication Diagram

Explanation

An Object Diagram represents instances of classes and their relationships at a particular moment in time. A Communication Diagram illustrates how objects exchange messages to accomplish specific system functions.

Explanation & Linkages

Object Diagrams provide the static view of a system, while Communication Diagrams represent its dynamic interactions. Together, they provide a complete understanding of object relationships and runtime collaboration.

Analysis

These UML diagrams improve system visualization, simplify software documentation, facilitate communication among stakeholders, and support effective object-oriented analysis and design.

(5 Marks)

6. Given:

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗

Identified Concepts

Coding Phase Failure, Implementation Errors

Explanation

Coding phase failure occurs when developers incorrectly implement the approved software design, resulting in logical, syntax, or runtime errors that prevent the application from functioning correctly.

Explanation & Linkages

Although requirement analysis and design may be completed successfully, improper coding introduces implementation defects that propagate into testing and deployment. Errors during this phase often originate from poor programming practices or inadequate code validation.

Analysis & Solution

Coding failures increase debugging effort, delay project completion, and reduce software quality. Following coding standards, conducting peer code reviews, and performing static code analysis significantly reduce implementation errors and improve software reliability.

(5 Marks)

7. Construct a Concept Map

Given:

Problem Identification → _____ → Design → Implementation → _____

Answer:

Identified Concepts

Requirement Analysis, Testing

Explanation

Requirement Analysis identifies user needs, system constraints, and project objectives. Testing verifies whether the implemented software satisfies these requirements and performs correctly under different operating conditions.

Explanation & Linkages

Problem-solving begins by understanding user expectations through requirement analysis. Design and implementation transform these requirements into software. Testing validates the final system against the original requirements, ensuring correctness and completeness.

Analysis

This structured approach minimizes development errors, improves software quality, reduces project risks, and increases the likelihood of successful software delivery.

(5 Marks)

8. Construct a Concept Map

Given:

Object-Oriented Features

→ Class

→ _____

→ Message Passing

→ _____

Answer:

Identified Concepts

Object, Dynamic Binding

Explanation

Dynamic Binding determines the method to execute during runtime rather than compile time, allowing objects to exhibit different behaviors through polymorphism.

Explanation & Linkages

Objects communicate through message passing. Dynamic binding ensures that the correct method implementation is selected according to the actual object type, enabling flexible and extensible software architectures.

Analysis

Dynamic binding supports runtime polymorphism, improves code flexibility, simplifies maintenance, and enables software systems to accommodate future enhancements without major structural changes.

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(5 Marks)

9. Construct a Concept Map

Given:

Software Development Models

→ Waterfall

→ _____

→ Spiral

→ _____

Answer:

Identified Concepts

Incremental Model, Agile Model

Explanation

The Incremental Model develops software in multiple functional releases, whereas the Agile Model focuses on iterative development, customer collaboration, and continuous improvement.

Explanation & Linkages

Incremental development divides large projects into manageable modules delivered in stages. Agile enhances this process through frequent customer feedback, adaptive planning, and rapid iterations, enabling teams to respond quickly to changing requirements.

Analysis

Both models reduce project risks, improve product quality, increase customer involvement, and provide greater flexibility than traditional sequential development approaches.

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(5 Marks)

10. Construct a Concept Map

Given:

Class Diagram

→ Class

→ _____

→ Association

→ _____

Answer:

Identified Concepts

Attribute, Method

Explanation

Attributes represent the data stored within a class, while methods define the operations performed using that data. Together, they describe the internal structure and functionality of an object.

Explanation & Linkages

Class diagrams organize software into logical units by representing classes, their attributes, methods, and relationships. Methods manipulate attribute values, allowing objects to perform meaningful operations while maintaining encapsulation.

Enhanced Analysis

Well-designed class diagrams improve software documentation, simplify implementation, support code reuse, and provide the structural foundation for object-oriented software systems.

(5 Marks)

11. Construct a Concept Map

Given:

Software Engineering Principles

→ Modularity

→ _____

→ Reusability

→ _____

Answer:

Identified Concepts

Maintainability, Scalability

Explanation

Maintainability refers to the ease of modifying software to fix defects or introduce new features. Scalability enables software to efficiently support increasing numbers of users, transactions, and data without performance degradation.

Explanation & Linkages

Maintainability is achieved through modular design, proper documentation, and reusable components. Scalability depends on efficient architecture, optimized algorithms, and flexible system design capable of handling future growth.

Analysis

Applying these principles reduces maintenance costs, improves software performance, supports business expansion, and extends the operational life of software systems.

(5 Marks)

12. Given:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ →
Deployment ✗

Identified Concepts

Integration Phase Failure, Module Communication

Explanation

Integration failure occurs when independently developed software modules fail to communicate or exchange data correctly after integration, despite functioning properly when tested individually.

Explanation & Linkages

Successful coding does not guarantee successful integration. Module incompatibilities, interface mismatches, API errors, and inconsistent data formats often cause failures during integration. Effective coordination between modules is essential before deployment.

Analysis & Solution

Integration failures delay software release, increase debugging complexity, and reduce system reliability. Performing comprehensive integration testing, validating interfaces, standardizing communication protocols, and resolving compatibility issues early ensure smooth system integration and successful deployment.

(5 Marks)

CONCEPT MAPPING

RUBRICS FOR CALCULATION

CONCEPT MAPPING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand